

Laurinne J Balstad
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Positions

2026- **University of British Columbia**
Postdoctoral researcher, King lab

Education

2025 **University of California, Davis**
PhD in Population Biology
Advisor: Marissa Baskett
Thesis: Facilitating, Promoting, and Impeding Evolutionary Rescue in the Anthropocene

2021 **St. Olaf College, Bachelor of Arts**
Mathematics and Biology, double major

Publications

- 8 **Balstad, Laurinne**, Baskett, Marissa. Drivers of evolutionary rescue in host-pathogen systems. *Accepted at PNAS*.
- 7 **Balstad, Laurinne***, Brennan, Joe*, [12 authors]. **2026**. Eight quick tips for data-model integration. *Accepted at PLoS Computational Biology*. *Shared first authorship.
- 6 Urban, Mark and **Balstad, Laurinne**. **2026**. *Perspective: Evolution to the rescue. Science* 391, 6790. [Invited]
- 5 **Balstad, Laurinne**, Baskett, Marissa, Carlson, Stephanie, Cunningham, Curry, & Westley, Peter. **2026**. Revisiting evolutionary rescue in the wild. *Trends in Ecology & Evolution*. 41(8), 762–774.
- 4 Szuwalski, Cody, Alglave, Baptiste, **Balstad, Laurinne**, Olmos, Maxime, Punt, Andre, Richar, Jon, Stockhausen, William, and Veron, Matthieu. **2026**. Density dependence modulates climate change impacts on eastern Bering Sea snow crab. *J. of Applied Ecology*, 63(3), e70316.
- 3 **Balstad, Laurinne**, Godwin, Sean, Krkosek, Martin, Lewis, Mark, and Baskett, Marissa. **2025**. Threshold-based disease treatment approach modulates economic, conservation and evolutionary trade-offs in sea louse-salmon aquaculture system. *Theoretical Ecology*, 18(1), 23.
- 2 **Balstad, Laurinne**, Fedewa, Erin, and Szuwalski, Cody. **2024**. Drivers of Bitter Crab Disease occurrence in Eastern Bering Sea snow crab (*Chionoecetes opilio*). *ICES J. of Marine Science*, 81(6), 1073–1083.
- 1 **Balstad, Laurinne**, Binning, Sandra, Craft, Meggan, Zuk, Marlene, and Shaw, Allison. **2021**. Parasite intensity and the evolution of migratory behavior. *Ecology*, e03229.

White papers

- 11 Kinneen, Michael, Goodman, Maurice. C., Sulc, Anna, **Balstad, Laurinne**, ... [19 authors]. **2025**. Status of widow rockfish stock off the U.S. West Coast in 2025. *Pacific Fishery Management Council [peer reviewed]*.
- 10 **Balstad, Laurinne**, Sullivan, Jane, and Monnahan, Cole. **2024**. A Plain Language Guide to Model Validation in REMA. White paper, *North Pacific Fisheries Management Council [peer reviewed]*.

- 9 Pinsky, Malin, ... **Balstad, Laurinne**, ... [17 authors]. **2024**. Evolutionary Resilience for Kelp: Prospects and Challenges. *Workshop Report, Monterey Bay Aquarium, University of California, Santa Cruz*. doi: 10.6084/m9.figshare.26882734

Manuscripts in the pipeline, available upon request

Balstad, Laurinne, Ashby, Ben, Godwin, Sean, Krkosek, Martin, Lewis, Mark, and Baskett, Marissa. Predation dynamics shape sublethal virulence evolution in macroparasites. *In prep.*

Jarvis-Cross, Madeline*, **Balstad, Laurinne***, Amy A. Briggs, Matthew Mensinger, Anjana Parandhaman, Eric Palkovacs, Marissa L. Baskett. Mitigating unintended consequences of population supplementation in imperiled host-pathogen systems. **Shared first authorship. Submitted to Conservation Science and Practice.*

Grants & Funding (\$235K USD)

- 2025 Jastro-Fields Small Grant, *University of California, Davis*, **\$1,500 Stipend**
 2025 Graduate Student Travel Award, *University of California, Davis*, **\$1,500**
 2024 Jastro-Fields Small Grant, *University of California, Davis*, **\$3,000 Undergraduate researcher stipend support**
 2023 Jastro-Fields Small Grant, *University of California, Davis*, **\$2,775 Fieldwork travel/fees**
 2021 Graduate Research Fellowship, *National Science Foundation*, **\$144,000**
 2021 Graduate Research Traineeship, *National Science Foundation, Sustainable Oceans NRT* **\$46,000**
 2021 Dean's Distinguished Graduate Fellowship, *University of California, Davis*, **\$39,000**

Teaching

- 2025 **Instructor of Record** (co-taught), Population Ecology, *University of California, Davis*
Certificate "Seminar on College Teaching" [10 weeks]
 2024 **Teaching assistant**, Population Ecology, *University of California, Davis*
Discussion leader and course organizer (TA), Marine disease (graduate course), *University of California, Davis*
 2023 **Teaching assistant**, Marine Ecosystem Research and Management, University of New Hampshire
Guest lectures: Data Management; Ecosystem indicators: Bitter crab disease case study
Discussion leader and course organizer (TA), Ecosystem-based fisheries management (graduate course), *University of California, Davis*

Student research mentorship

- 2026 Post-baccalaureate researcher, "Predicting shifts in emergence hotspots of avian influenza in arctic marine mammals under climate change"
 seed2STEM High school student researcher, "Marine disease in fisheries management"
Program to provide research experience to Indigenous high school students
 2025 Undergraduate researcher, "Reintroductions and evolutionary rescue in host-pathogen systems"
 2023 Undergraduate researcher, "Patterns of cohort resonance across Pacific coast fish stocks"

Presentations

- Upcoming *Invited seminar*, University of Alaska, Fairbanks, Center for Fisheries and Ocean Science Seminar Series

- Invited seminar*, Maud Menten Institute of Mathematics for the Life Sciences (hosted at University of Manitoba), Maud Menten Institute Seminar Series
- Invited seminar*, University of Victoria, Mathematical Biology Seminar Series
- 2026 *Seminar*, University of British Columbia Biodiversity Internal Seminar Series
- Talk*, University of British Columbia, Mathematical Biology Symposium Fall 2026
- Poster*, Ecological Society of America Annual Meeting (Salt Lake City, US)
- co-Session Organizer ("Vector-borne disease")* at Ecological Society of America Annual Meeting (Salt Lake City, US)
- Invited attendee*, NAS-Royal Society Forum on Shifts and tipping points in ocean systems (London, UK)
- Talk*, University of British Columbia, Mathematical Biology Symposium Spring 2026
- Talk*, Society of Mathematical Biology, Population Dynamics Conference (Online)
- 2025 *Invited panelist*, University of California, Davis, One Health Symposium
- Seminar*, University of California, Davis, Center for Population Biology Seminar Series
- Talk & co-Session Organizer ("Evolutionary Rescue")*, Ecological Society of America Annual Meeting (Baltimore, US)
- Talk*, Max Planck Evolutionary Rescue Workshop (Plön, Germany)
- 2024 *Talk*, Western Society of Naturalists (Portland, US)
- Seminar*, University of California, Davis, Center for Population Biology Seminar Series
- 2023 *Poster*, Western Society of Naturalists (Monterey, US)
- Talk*, Alaska Marine Science Symposium (Anchorage, US)
- 2021 *Talk*, SIAM Dynamical Systems (Online)
- 2020 *Poster*, Society of Mathematical Biology Annual Meeting (Online)

Service

- 2026 **Vice Chair, Disease Ecology Section**, Ecological Society of America
- 2025 **Workshop co-leader**, Introduction to R and poster making (9 lesson series), *University of California, Davis Ecology, Evolution, and Working Landscapes REU*
- 2025 **Graduate Student Representative Center for Population Biology Postdoctoral Fellowship Committee**, University of California, Davis
- 2022-2025 **Co-chair and program leader, Ecology & Evolution Preview Program**, University of California, Davis, *including Co-chair, Mentorship co-coordinator, Workshop presenter*
Program encourages students (n > 150) from underrepresented backgrounds to consider and apply to graduate school in Ecology & Evolution by making application process transparent
- 2021-2025 **Student Diversity Committee**, University of California, Davis
- 2024 **Professors for the Future**, *University of California, Davis*
Professional development program, with service project "Chat with and editor: Understanding the hidden curriculum of publishing" (workshop series)
- Reviewer *Evolutionary Applications, American Naturalist, Ecosphere, Ecological Applications*

Outreach

- 2024 **NOAA**: Modeling Science Bioenergetics for Grades 6-8
Collaboration with Alaska Fisheries Science Center, NOAA
Shinyapp with lesson plan linked: https://lauriebalstad.shinyapps.io/SAL_GROW/
- University of Minnesota**: Simulating selection on migration for Grade 12
Collaboration with Martha Torstenson and Allison Shaw
Lesson plan with Shinyapp: <https://qubeshub.org/publications/5103/2>